

TED PAULO A. FERANIL

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EDUCATION

Polytechnic University of the Philippines – Manila

Manila, PH

Bachelor of Science in Computer Engineering Major in AI/ML

2022 – Present (Expected August 2026)

Thesis title: EarPrompt: A Real-Time Conversational AI Assistant for Enhancing Verbal Communication

Awards: DOST Scholar (National Scholarship), Consistent President Lister (Top Academic Honor)

EXPERIENCE

Department of Science and Technology – PES - ITD

Manila, PH

Programmer/Developer Intern

July 2025 – September 2025

PHP | CSS | HTML | JavaScript | Laravel | Bootstrap | MySQL

- Designed a full-stack development of an internal stock management platform for DOST.
- Implemented 15+ database migrations, core CRUD operations for inventory management, and UI components, ensuring system stability through debugging and performance tuning alongside senior developers.

NTEK Systems Inc.

Manila, PH

AI Software Developer Intern

July 2024 – September 2024

Python | LangChain | FAISS | Flask | LLM Integration

- Engineered a Retrieval-Augmented Generation (RAG) pipeline linking LLM to documents, enabling real-time document querying and summarization across 20+ internal files.
- Created Flask-based backend APIs supporting the pipeline, improved query response time by ~70% (from ~8s to 2–3s) via iterative retrieval and optimized indexing.

PROJECTS

EarPrompt: Real-Time Conversational AI Assistant

Python | Flask | LLM Integration | Flutter | Firebase | Raspberry Pi 5

- Architected and led backend of a real-time voice AI pipeline, integrating Praat for voice analysis, SpeechBrain for speech recognition, and ElevenLabs for TTS; Flutter mobile app as the control interface.
- Ran the pipeline on Raspberry Pi 5, keeping end-to-end latency < 2 s; rated High(2.83/4) by speech pathologists.

SupportIQ (RAG-Based Company Support Chatbot)

Python | FastAPI | LangChain | Supabase | React | Tailwind

- Built an end-to-end RAG pipeline with recursive text splitting, vector embedding, and hybrid semantic + keyword search with similarity-threshold fallback for improved retrieval accuracy.
- Developed and deployed a FastAPI +React on Vercel and Render, tuning LLM parameters to produce grounded, non-repetitive responses.

Banana Ripeness Classifier (SVM)

Python | Scikit-learn | OpenCV | SVM | Image Processing | Streamlit

- Trained a multi-class SVM model to classify banana ripeness (unripe, ripe, overripe, rotten) from image color features using Scikit-learn.
- Preprocessed image data with OpenCV and evaluated model performance across 4 classes achieving 93% accuracy.

SKILLS & INTERESTS

Languages: Python, PHP, JavaScript, HTML, CSS

Frameworks & Libraries: Flask, FastAPI, Laravel, Bootstrap, Tailwind, LangChain

AI & Data: RAG Pipeline, Prompt Engineering, Hugging Face, Scikit-learn

Databases: MySQL, Supabase, Firebase (Firestore)

Tools: Git, GitHub, VS Code, Postman, REST APIs

Interests: Building AI-powered applications, AI agent systems